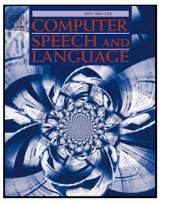
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Incorporating external knowledge for text matching model

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ABSTRACT

Text matching is a computational task that involves comparing and establishing the semantic relationship between two textual inputs. The prevailing approach in text matching entails the computation of textual representations or employing attention mechanisms to facilitate interaction with the text. These techniques have demonstrated notable efficacy in various text-matching scenarios. However, these methods primarily focus on modeling the sentence pairs themselves and rarely incorporate additional information to enrich the models. In this study, we address the challenge of text matching in natural language processing by proposing a novel approach that leverages external knowledge sources, namely Wiktionary for word definitions and a knowledge graph for text triplet information. Unlike conventional methods that primarily rely on textual representations and attention mechanisms, our approach enhances semantic understanding by integrating relevant external information. We introduce a fusion module to amalgamate the semantic insights derived from the text and the external knowledge. Our methodology's efficacy is evidenced through comprehensive experiments conducted on diverse datasets, encompassing natural language inference, text classification, and medical natural language inference. The results unequivocally indicate a significant enhancement in model performance, underscoring the effectiveness of incorporating external knowledge into text-matching tasks.

1. Introduction

Text matching is a fundamental research area in natural language processing (Jiang et al., 2022a). It plays a crucial role in measuring the semantic or intentional similarity between two texts and is considered a core problem in the field of NLP. Text matching serves as a foundational technique for various tasks and has proven successful in domains such as reading comprehension (Liu et al., 2023), question-answering systems (Behmanesh et al., 2023), and machine translation (Kang et al., 2023).

In machine reading comprehension tasks, semantic matching between the context and the question is utilized to identify the correct answer. For question-answering systems, sentence matching is vital in two key aspects: (1) question retrieval, which determines the similarity between queries and existing questions, and (2) answer selection, which assesses the extent of correspondence between queries and answers. In machine translation tasks, semantic matching between two languages can be used to evaluate the quality of the translation.

From Fig. 1, we can see that text-matching tasks encompass a range of objectives, including natural language inference, paraphrase identification, and answer selection. Textual entailment recognition focuses on capturing the relationship between premises and hypotheses, including instances of entailment, contradiction, and neutrality. Paraphrase identification, also referred

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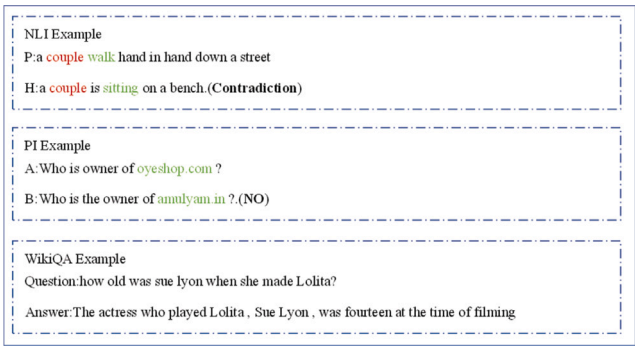


Fig. 1. Dataset case.

to as paraphrase detection, aims to identify pairs of sentences that convey the same meaning. Answer selection involves ranking candidate answers based on their matching scores with a given question.

The current text-matching models can be broadly divided into two categories: traditional methods and deep learning methods. Traditional methods heavily rely on manually defined features, making feature extraction difficult and leading to lower accuracy (Hu et al., 2020; Sultan et al., 2015). In recent years, with the rapid development of deep learning and the release of large-scale datasets such as SNLI, Quora, and SST, deep learning methods have become the mainstream approach. Deep learning methods can automatically extract text features and have high accuracy, making them widely applicable in natural language understanding tasks (Shen et al., 2020; Sugawara et al., 2020; Mittal et al., 2021), In 2018, with the introduction of the pre-trained language model BERT (Devlin et al., 2019), natural language understanding tasks entered a new era. BERT, which uses Transformer as an encoder, was pre-trained on large datasets and then fine-tuned on downstream tasks, achieving state-of-the-art results in text-matching tasks.

The aforementioned model has demonstrated remarkable achievements in the realm of text matching. However, it still exhibits several shortcomings that need to be addressed. Firstly, the models lack deep reasoning abilities and struggle with generalization. Although they excel in refining and complicating shallow-level matching, their capacity for deep-level inference between texts remains limited. Furthermore, most models tend to excel in only one or two tasks, failing to consistently deliver exceptional performance across multiple tasks and datasets. Secondly, the models rarely integrate external knowledge, leading to suboptimal outcomes. By relying solely on their internal learned knowledge, these models lack comprehensiveness. Therefore, incorporating external knowledge becomes crucial in facilitating more informed inferences. For instance, the inclusion of common-sense knowledge can greatly enhance logical reasoning between texts. Lastly, the models suffer from a lack of practical applications. Their current utility is constrained within specific domains, such as providing support to medical professionals in making judgments within the medical field.

To address the aforementioned issues, this study focuses on the following main research areas. (a) In response to the lack of deep reasoning abilities and weak capacity for deep-level inference between texts in existing models, resulting in low accuracy and poor generalization across multiple text-matching tasks, this study proposes a text-matching model based on multi-turn interactions and multi-feature integration. The model utilizes a deep encoder that incorporates multi-turn interactions and multi-feature integration. Compared to existing models, this encoder considers the internal information between texts. Additionally, a deep aggregation algorithm is employed to fuse the semantic information of the texts. After obtaining the text representations, a bidirectional attention mechanism is applied to enable a more profound semantic information exchange between the two texts. This ensures that the model maintains high accuracy while possessing robust generalization capabilities. (b) Incorporating External Knowledge into the Model: Addressing the issue of limited integration of external knowledge in existing models, this study retrieves relevant knowledge from external knowledge bases such as Wiktionary and ConceptNet (Lv et al., 2020). The retrieved knowledge is then encoded into knowledge vectors, which are introduced into the model using a gate mechanism to control the proportion of knowledge incorporation. (c) Application of Text Matching Models: The proposed model will be applied to sentiment analysis and medical inference tasks for experimental validation. Specifically, the main contributions of this paper as follows:

- We propose a novel text-matching model that leverages deep interaction and knowledge enhancement(DIKE). Our model employs multi-turn interactions to extract in-depth semantic information and incorporates Wiktionary and knowledge graphs to enhance its knowledge base. It aims to capture both the semantic information between text pairs and the semantic details of individual words, effectively integrating them using a gate mechanism.

- Our proposed method offers versatility, enabling its application to other models like ESIM and BERT. Furthermore, our approach demonstrates improved performance in sentiment analysis and medical reasoning tasks, highlighting its strong generalization capabilities.

- We evaluate our method through experiments on diverse datasets, including SNLI, Quora, Scitail, SICK, BreakNLI, as well as sentiment analysis dataset SST and medical reasoning dataset Mednli. The experimental results consistently validate the effectiveness of our approach across these datasets, underscoring its performance.

2. Relate work

2.1. Representation-based text matching model

Representation-based text matching methods typically involve encoding input text pairs into text vectors using an encoder and subsequently matching the two text vectors. In this process, the input text pairs are initially passed through an embedding layer to generate word vectors for each text. These word vectors are then fed into a representation layer, which typically consists of a text encoding layer and a pooling layer. For instance, Bowman et al. (2015) utilized an LSTM sentence model for text matching, where the two text segments are modeled with LSTM, followed by three activation layers. The concatenation of the resulting vectors is then input into an MLP. Facebook researchers proposed the Infersent dual-encoder model. Conneau et al. (2017) for sentence embeddings, utilizing various encoding methods in the representation layer. Through comparing different CNN and RNN-based encoders, they found that combining BiLSTM with max-pooling yielded the best performance. Building upon the dual-encoder architecture, Reimers and Gurevych (2019) introduced the SBERT model, which utilizes BERT for sentence encoding. Initially, the pooled output of BERT was used, but it performed poorly. Consequently, the authors fused the two vectors from multiple perspectives after obtaining the text vectors, resulting in significant improvements over directly concatenating the outputs of BERT. Another model addressing sentence embedding is SimCSE, proposed by Gao et al. (2021). This model employs BERT to obtain text vectors and utilizes contrastive learning to increase the distance between unrelated samples. SimCSE achieved state-of-the-art results on multiple datasets. While representation-based text matching methods primarily focus on encoding texts to obtain text vectors, emphasizing the information within the texts, they often neglect the interaction between texts, leading to suboptimal performance. Consequently, many researchers have shifted their focus toward studying the interaction between texts. Peng et al. (2023) proposed a dual-channel feature discrimination and fusion network (FDF). The FDF network utilizes a PLM-based encoder to extract features from both unabridged texts and abridged texts obtained through data deletion. Building upon this foundation, inter-channel gated feature fusion and interaction are employed to expand the textual representation with discerning and distinguishable features. The FDF network demonstrates outstanding performance on the MRPC, QQP, and BQ datasets.

Representation-based text matching methods focus on converting texts into vector representations, utilizing encoding layers such as LSTMs, CNNs, and advanced models like BERT. These approaches employ word embeddings and pooling strategies to effectively capture the essence of text pairs, creating comprehensive text representations. While these methods excel in constructing text vectors, they often overlook subtle text interactions, potentially leading to suboptimal performance in matching tasks. As understanding of textual interactions deepens, these methods are evolving towards a greater emphasis on the complex relationships between texts.

2.2. Interaction-based text matching model

Interaction-based text-matching models are typically composed of embedding layers, encoding layers, interaction layers, aggregation layers, and output layers. These models emphasize capturing complex interaction information between texts. Chen et al. (2017) proposed the ESIM model, which utilized a self-attentive mechanism and a bidirectional LSTM to encode and extract features from sentence pairs. Yang et al. (2019) proposed a text-matching model that performed multiple interactions using the original encoding vector, resulting in significant improvement. Deng et al. (2022) proposed a short text-matching model called Distance-Sensitive Self-Attention and Multi-angle Matching (DSSTM). The model integrates distance information between words in short texts into the self-attention mechanism, allowing attention to focus on closer words. Additionally, an interactive attention mechanism is employed to capture the interactions between short texts. Finally, a multi-angle similarity modeling approach is utilized to further enhance the effectiveness of short text matching in DSSTM. Lu et al. (2022) proposed a long text-matching model based on Aggregated Keyword Graph and Multi-head Graph Attention Network, The model introduces a method of modeling two long texts as Aggregated Keyword Graphs, which captures the spatial structure of text pairs. The model embeds local matching features of long text pairs into the Aggregated Keyword Graphs. Finally, the Multi-head Graph Attention Network is utilized to extract features from the Aggregated Keyword Graphs, further enhancing the matching performance of long text pairs. The model achieved F1 scores of 82.4% and 89.8% on the CNSE and CNSS benchmark datasets for long text matching, respectively, outperforming existing state-of-the-art models. Tang et al. (2022) proposed a multi-semantic interactive sentence matching model, named Multi-Semantic Interactive Model (MSIM), to address the challenge of Chinese short text matching. The model leverages multi-granularity semantic embeddings, including character embeddings based on one-hot vectors and word embeddings based on pre-trained models, to represent each sentence pair. The encoded sentences are then subjected to matching operations at each time step, followed by average pooling and max pooling. Deep-level interactions are facilitated by computing the difference and element-wise product between the attention representations and each time step. The model exhibits promising performance on the CCKS and ATEC datasets.

Interaction-based text matching models enhance matching precision by emphasizing complex interactions between texts, employing a multi-layer architecture that includes interaction and aggregation layers. These models leverage innovative techniques such as self-attention and distance-sensitive attention to focus on relationships within texts and interactions between text pairs, thereby improving feature extraction and enriching text representations. The evolution of these methods signifies a shift towards more contextually aware and semantically rich text-matching techniques, setting new performance benchmarks for matching tasks.

2.3. Pretrained-based text matching model

The emergence of the BERT (Devlin et al., 2019) model marked the onset of the pre-trained language model era in natural language processing (NLP), it was constructed by stacking multiple layers of Transformers, revolutionizing NLP with its powerful capabilities. During the pre-training phase, BERT tackles two tasks: (1) Masked Language Model (MLM), which resembles a cloze test, randomly masking 15% of words and predicting them based on contextual cues. Among the 15% masked words, 80% is replaced with the "[MASK]" token, 10% are substituted with other words, and 10% remains unchanged. (2) Next Sentence Prediction (NSP), which determines if a second sentence follows the first sentence, serving tasks that require sentence-level representations like question answering and semantic similarity. BERT's exceptional feature extraction abilities outperform prior methods on various text-matching datasets. Liu et al. (2019) conducted an analysis of BERT and identified that the second pre-training task, NSP, does not contribute significantly to its performance. Building upon this insight, they introduced the RoBERTa model. RoBERTa eliminates the NSP task and incorporates a dynamic masking mechanism, achieving superior performance to BERT on training data. Therefore, RoBERTa surpasses BERT in terms of overall effectiveness. While BERT and RoBERTa focus on natural language understanding, the OpenAI team proposed the GPT (Generative Pre-trained Transformer) series (Dehouche, 2021). Similar to BERT, GPT models employ a stack of Transformer layers in the decoder configuration, enabling excellent text generation capabilities. Raffel et al. (2020) proposed the T5 model, which formulates all NLP tasks as generation tasks, utilizing the complete Transformer architecture with relative position encoding. T5 achieves state-of-the-art performance on multiple datasets. Although these pre-trained language models have demonstrated significant advancements in various NLP tasks, their large parameter sizes pose challenges. To address this, Jiao et al. (2020) proposed the TinyBERT model, employing knowledge distillation to transfer language knowledge from the BERT model to a smaller variant, TinyBERT. Experimental results reveal that TinyBERT achieves comparable performance while providing a 9.4x speedup and reducing the model size by 7.5x. Additionally, Lan et al. (2019) introduced ALBERT, a simplified version of BERT. ALBERT incorporates two parameter reduction techniques: (1) Factorized Embedding Parameterization, decomposing the word embedding matrix into smaller matrices to separate embedding and hidden layer dimensions, and (2) Cross-Layer Parameter Sharing, enabling parameter sharing across different layers. Compared to BERT-large, ALBERT reduces parameters by 1/18 and trains 1.7x faster. These pre-trained language models have yielded significant advancements in diverse NLP tasks, including text matching and beyond.

Pre-trained language model-based text matching methods leverage models pre-trained on large corpora, such as BERT and GPT, to capture the deep semantic information of texts. By fine-tuning these pre-trained models for specific matching tasks, these methods can comprehend contextual meanings of texts, enhancing matching accuracy and efficiency. With the continuous improvement of pre-trained models in processing natural language, these approaches offer a powerful and flexible solution for text matching, enabling efficient matching across a wider range of applications.

2.4. Knowledge enhanced-based text matching model

In order to enhance the performance of models, researchers have also incorporated external knowledge into text-matching models. Chen et al. (2018) used external knowledge to enrich the natural neural language inference model KIM. which incorporates artificially constructed knowledge like WordNet. It achieves 88.6% accuracy on the Snli dataset by incorporating external knowledge such as synonyms and antonyms from Wordnet and achieves a 0.6% improvement compared to the ESIM model without the knowledge. Wang et al. (2019) considered that most textual inference models use only textual information and do not make good use of the diverse inference information present in open domain knowledge graphs. Therefore, the ConSeqNet model is proposed to improve the model performance by exploiting external knowledge information. The structured knowledge is incorporated into the text model by constructing a graph model, and good results are achieved. Gajbhiye et al. (2021) proposed ExBERT model, which transforms knowledge triads into textual descriptions and then inputs them into BERT, has greatly improved experimental results on multiple datasets compared to the basic BERT model. Yu et al. (2023) proposed SEIN, a lightweight text-matching model that integrates external knowledge from the WordNet database to enrich textual semantic information. The model facilitates attention-based interactions between the retrieved external knowledge and the textual information, thereby achieving optimal performance across various datasets.

Knowledge-enhanced text matching methods enrich the semantic layers of text matching by integrating external knowledge bases or knowledge graphs. These methods rely not only on the information within the texts themselves but also utilize domain-specific knowledge to improve the accuracy and depth of matching. By incorporating entities and relationships from knowledge graphs into the text-matching process, these approaches can understand and match texts at a deeper semantic level, offering more precise and comprehensive solutions for complex matching tasks.

3. Methodology

Text matching represents a fundamental problem in natural language processing (NLP), with the following formal definition: Given a labeled training set $D_1 = \left\{ \left(s_1^{(i)}, s_2^{(i)} \right), r^{(i)} \right\}_{i=1}^{N}$, where $s1^{(i)} \in S_1$ and $s_2^{(i)} \in S_2$ are two text segments in the training set, and $r^{(i)}$ denotes the relationship between these two segments. The objective of text matching is to learn a mapping function $f : S_1 \times S_2 \longrightarrow R$ that accurately predicts the relationship $r^{(i)}$ for any given text inputs $s_1^{(i)} \in S_1$ and $s_2^{(i)} \in S_2$ in the test set D_2. This can be represented as Eq. (1):

$$Match\left(s_1^{(i)}, s_2^{(i)}\right) = F\left(\psi\left(s_1^{(i)}\right), \psi\left(s_2^{(i)}\right)\right) \tag{1}$$

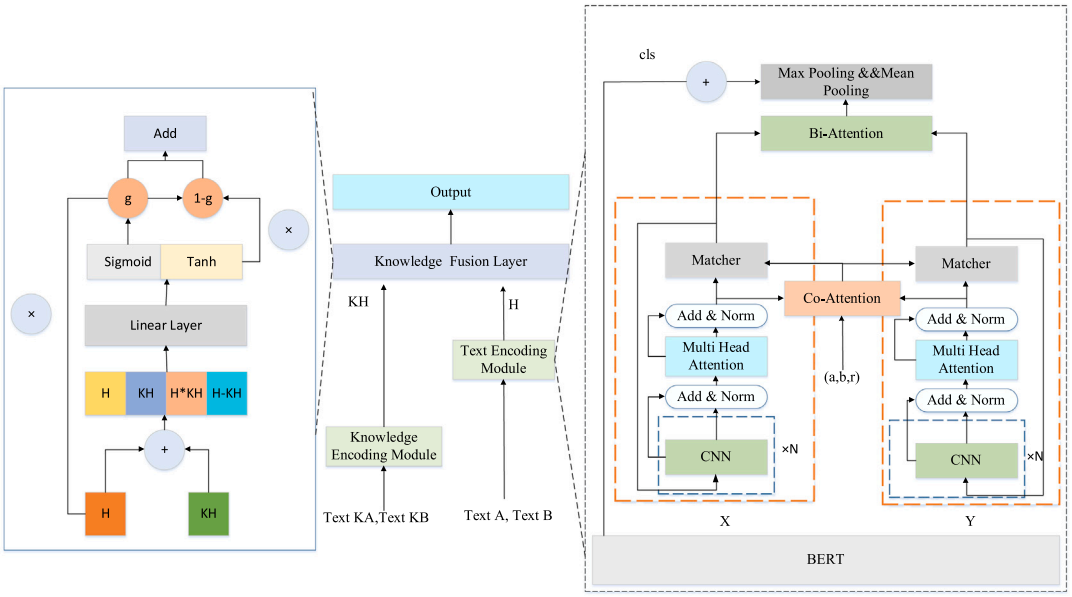


Fig. 2. Overview of the architecture of our proposed model. The middle shows the overall model framework proposed in this paper, the left side shows the flow chart of the knowledge fusion layer, and the right side shows the text encoding module proposed in this paper, which can be an arbitrary text matching.

Source text	P:The man is holding a saxophone H: The man is holding an instrument
Wiktionary	`saxophone': 'A single-reed instrument musical instrument of the woodwind family, usually made of brass and with a distinctive loop bringing the bell upwards' `instrument': 'A device used to produce music'.
Conceptnet	saxophone IsA instrument

Fig. 3. Premise (P), Hypothesis (H), and Knowledge from Conceptnet and Wiktionary.

where, $s_1^{(i)}$ refers to the first text segment in the ith text pair, $s_2^{(i)}$ refers to the second text segment in the ith text pair, ψ denotes the function that converts text into vectors, and F represents the text matching function.

In this subsection, we will describe our model in detail. As shown in Fig. 2, The deep interaction and knowledge enhancement(DIKE) model is mainly composed of the knowledge encoding layer, text encoding layer, knowledge fusion layer, and output layer. The following text will provide a detailed introduction to each layer.

3.1. Knowledge encoding module

In this section, we will introduce our knowledge retrieval approach and describe the method for encoding knowledge.

3.1.1. Knowledge retrieval

Knowledge Retrieval is to retrieve relevant and useful information from massive unstructured and messy knowledge base texts. As shown in Fig. 3, it is an example of knowledge enhancement, where we selected a sample from the SNLI dataset, and the red part represents the added external knowledge. From Fig. 3, it can be seen that without the addition of knowledge, the model has difficulty inferring that the saxophone is an instrument. However, adding the definition of saxophone, which indicates that it is a musical instrument, enables the model to correctly reason about their relationship.

We use ConceptNet (Lv et al., 2020) knowledge graph and Wiktionary (Xu et al., 2021) as external knowledge sources and adopt word definitions as external knowledge. Wiktionary contains 999,614 entity descriptions. For each entity, we choose the first entity description in Wiktionary as our external knowledge. As shown in Fig. 4, it illustrates the whole process of our knowledge retrieval. First, we retrieve relevant words from the nodes of the ConceptNet knowledge graph for the data we want to process, with the aim of extracting meaningful entity vocabulary. For each entity, we select the first entity description in Wiktionary as our external knowledge. To avoid some vocabulary not finding the correct form in Wiktionary, we use the lemma form of the word. For example, the word "singing" is not in Wiktionary, but the word "sing" is described as "To produce musical or harmonious sounds with one's

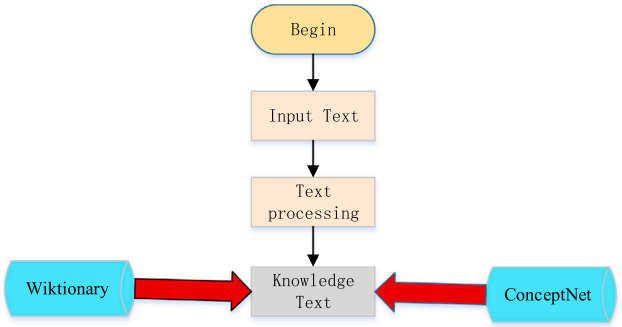


Fig. 4. Knowledge retrieval process.

voice" in Wiktionary. Therefore, we can convert "singing" to "sing" and find it in Wiktionary. Through this process, We can retrieve the definition texts KA and KB for the texts A and B respectively, as well as the relationship r between their words.

3.1.2. Knowledge encoding

For the knowledge text we retrieved, we input it into BERT that is contains L layers transformers and MLP models to obtain the knowledge representation vectors, the process is shown in Eq. (2):

$$\begin{aligned} h^L, H^L &= Transformer(KA, KB) \\ KH &= MLP\left(h^L\right) \end{aligned} \tag{2}$$

where h^L represents the representation of the first token "[CLS]", H^L represents the representation of the entire input of the Lth layer of the Transformer

3.2. Text encoding module

In this section, we need to encode the text. We can see this from the right side of Fig. 2. The text encoding module is mainly composed of the following layers, which are divided into the encoding layer, cross-attention layer, aggregation layer, bidirectional attention layer, and pooling layer.

3.2.1. Encoder layer

We given two input texts $A = \left(a_1, a_2, \ldots, a_n\right)$ and $B = \left(b_1, b_2, \ldots, b_m\right)$, where a_i and b_j are the tokens for each texts. We input them separately into multiple BERT models with shared parameters, obtaining their overall encoding information $[cls]$ as well as their individual encoding information X and Y.

After obtaining the encoded representation X for sentence A and Y for sentence B through the encoding layer, in order to extract more detailed encoding information, this article first uses multiple CNNs to extract local features and then uses multi-head self-attention to obtain global features. We take the encoding of text A as an example, which is shown in Eq. (3):

$$\begin{aligned} \mathbf{A_c} &= Conv(\mathbf{X}) \\ \mathbf{A}_m &= MultiHead\left(\left[\mathbf{A_c}; \mathbf{X}\right]\right) \\ \mathbf{M} &= \left[\mathbf{A_c}; \mathbf{A_m}\right] \in R^{n*d}. \end{aligned} \tag{3}$$

where $\mathbf{A_c}$ is the result of one-dimensional convolution, $\mathbf{A_m}$ is the result of multi-head attention mechanism, $\left[\mathbf{A_c}; \mathbf{X}\right]$ and $\left[\mathbf{A_c}; \mathbf{A_m}\right]$ are concatenation operations. Similarly, we can obtain the encoding of text B as $\mathbf{N} \in R^{m*d}$.

3.2.2. Cross attention layer

We referred to the approach of Jiang et al. (2022b) in this module to obtain the text representation M' and N'. However, we differ in that we incorporated the ConceptNet triplet information into the similarity matrix. Specifically, we increase the similarity value when there is a relation between the words in the two texts in ConceptNet. Which is shown in Eq. (4):

$$\begin{aligned} S &= f(M, N) + I \\ I_{ij} &= \begin{cases} 1 & \text{if } \exists(a_i, b_j) \\ 0 & \text{else} \end{cases} \end{aligned} \tag{4}$$

where f is a function that calculates the similarity.

3.2.3. Aggregation layer

The purpose of the aggregation layer is to aggregate the textual information before and after cross-attention. Eq. (5) demonstrates how the aggregation layer combines the three perspectives before and after cross-attention to obtain a final representation for text A. The operators used in this process include $\odot$ for element-wise multiplication, subtraction for vector difference, and multiplication for vector similarity. The aggregation layer employs four forward neural networks, denoted as G_1, G_2, G_3, and G, to compute the intermediate representation U, respectively. Similarly, the representation V for text B can be obtained using the same process.

$$\begin{aligned} a_1 &= G_1\left(\left[M; M'\right]\right) \\ a_2 &= G_2\left(\left[M; M - M'\right]\right) \\ a_3 &= G_3\left(\left[M; M \odot M'\right]\right) \\ U &= G\left(\left[a_1; a_2; a_3\right]\right) \end{aligned} \tag{5}$$

3.2.4. Bidirectional attention layer

The purpose of the bidirectional attention layer is to enable the text information we obtain to interact again, in order to achieve a deeper level of interaction. Our method involves utilizing the technique presented by Jiang et al. (2022a) for computing the attention Q from U to V and the attention C from V to U. The attention originates from the similarity matrix $\boldsymbol{T}$, where t_{ij} denotes the similarity between the ith word of U and the jth word of V.

$U \rightarrow V$: The attention describes which words in the text U are most relevant to V. The calculation process is similar to that of the co-attention layer, We can obtain the attention matrix $T \in R^{d*n}$, which is calculated as shown in Eq. (6).

$$\begin{aligned} \alpha_t &= softmax(T_{t:}) \in R^n \\ C_{:t} &= \sum_j \alpha_{tj} V_{:j} \end{aligned} \tag{6}$$

where $T_{t:}$ is the tth raw of T, and $c_{:t}$ is the tth column of C.

$V \rightarrow U$: The attention indicates which words in V are most similar to U. The calculation process is as follows: first, the column with the largest value in the similarity matrix $\boldsymbol{T}$ is taken to obtain the attention weight, then the weighted sum of C is expanded by n time steps to obtain $V \in R^{d*n}$, which is calculated as shown in Eq. (7).

$$\begin{aligned} b &= softmax(\max_{col}(T)) \in R^m \\ q &= \sum_t b_t U_{t:} \in R^d \end{aligned} \tag{7}$$

Once Q and C are obtained, we merge the attention from both directions using a multilayer perceptron, resulting in the contextual representation $G \in R^{4d*n}$ as shown in Eq. (8).

$$\begin{aligned} \mathbf{G}_{:\mathbf{t}} &= \beta\left(\mathbf{C}_{:\mathbf{t}}, \mathbf{V}_{:\mathbf{t}}, \mathbf{Q}_{:\mathbf{t}}\right) \\ \beta(\mathbf{c}, \mathbf{v}, \mathbf{q}) &= \left[\mathbf{v}; \mathbf{q}; \mathbf{v} \odot \mathbf{q}; \mathbf{v} \odot \mathbf{c}\right] \in R^{4d} \end{aligned} \tag{8}$$

3.2.5. Pooling layer

The aim of the pooling layer is to extract important information from the text. This paper utilizes both average pooling and maximum pooling techniques, and their results are concatenated directly. This calculation is expressed in Eq. (9).

$$\begin{aligned} \mathbf{G}_{\max} &= MAX(\mathbf{G}) \\ \mathbf{G}_{\mathbf{mean}} &= MEAN(\mathbf{G}) \\ \mathbf{O} &= \left[\mathbf{G}_{\max}; \mathbf{G}_{\mathbf{mean}}\right] \end{aligned} \tag{9}$$

where, MAX represents max pooling, $MEAN$ represents mean pooling, and $\mathbf{O}$ represents the pooled vector. Then, we use an MLP network to obtain text features H from the global vector '[cls]' and the pooled vector.

3.3. Knowledge fusion module

After passing through the knowledge encoding module and the text encoding module, we obtain the text representation $\mathbf{H}$ and the knowledge representation $\mathbf{KH}$. In order to effectively integrate the information of the two and reduce the influence of noise, we first perform a differential calculation on the text representation $\mathbf{H}$ and the knowledge representation $\mathbf{KH}$, i.e., we calculate their inner product $\mathbf{H} \odot \mathbf{KH}$ and difference $\mathbf{H} - \mathbf{KH}$, and concatenate them with $\mathbf{H}$ and $\mathbf{KH}$, so that we can obtain the relationship and difference between the text and knowledge. Then, we use a gating mechanism to perform fusion, that is, using a neural network to control the fusion ratio of them, and obtain the final output $\mathbf{Z} = fusion(\mathbf{H}, \mathbf{KH})$, where the fusion function $Z = fusion(x, y)$ is defined as Eq. (10):

$$\begin{aligned} \tilde{x} &= \tanh\left(W_1\left[x; y; x \odot y; x - y\right]\right) \\ g &= sigmoid\left(W_2\left[x; y; x \odot y; x - y\right]\right) \\ z &= g \odot \tilde{x} + (1 - g) \odot x \end{aligned} \tag{10}$$

where, $\mathbf{W_1}$ and $\mathbf{W_2}$ are weight matrices, and g is a gating mechanism that controls the proportion of the intermediate vector in the output vector.

3.4. Output module

The purpose of the output module is to output its corresponding result. In this article, a multilayer perceptron is used to obtain the result of text matching. The process is shown in Eq. (11):

$$y = softmax(\tanh(\mathbf{ZW} + \mathbf{b})) \tag{11}$$

where, $\mathbf{W}$ and $\mathbf{b}$ are both trained parameters.

Finally, the cross-entropy loss function is used to calculate the loss value, which is calculated as follows:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{i,j}\log\left(p_{i,j}\right) \tag{12}$$

where N is the number of samples, C is the total number of categories, and y_i is the true label of the ith sample.

4. Experiments

In this chapter, we first introduce some implementation details and the baseline of the model. Then, we present the experimental results on the dataset. Finally, we analyze the experimental results.

4.1. Dataset

We conducted experiments using text-matching datasets SNLI, Scitail, Quora, and SICK, text classification datasets SST-5, and medical natural language inference dataset MedNLI.

SNLI (Bowman et al., 2015): SNLI (Stanford Natural Language Inference) is a large publicly available dataset that is designed for natural language inference tasks. It was released by the Stanford University AI Lab and consists of 570,000 sentence pairs, which are divided into training, development, and test sets. Each sentence pair contains a premise and a hypothesis, both of which are natural language texts. The premise is a descriptive sentence, and the hypothesis is a statement that can be inferred from the premise.

Scitail (Khot et al., 2018): Scitail is a publicly available dataset that is designed for natural language inference tasks, with a specific focus on reasoning problems in the fields of science and technology. It was released by the Allen Institute for Artificial Intelligence and contains 27,026 sentence pairs, which are divided into training, development, and test sets. Each sentence pair comprises a premise and a hypothesis, both of which are natural language texts.

SICK (Marelli et al., 2014): SICK (Sentences Involving Compositional Knowledge) is a publicly available dataset designed for natural language inference tasks. The dataset was released by researchers from the University of Stuttgart in Germany. The SICK dataset comprises 9841 sentence pairs that are divided into training, development, and test sets. Each pair consists of a premise and a hypothesis, both of which are natural language texts. The task involves judging whether the hypothesis is entailed by the premise, contradicts the premise, or is neutral, based on the relationship between the premise and the hypothesis.

Quora (Aghaebrahimian, 2017): The Quora dataset is a publicly available dataset specifically designed for question–answer matching tasks. The dataset was released by the Quora community with the aim of promoting research and development in the fields of question–answer systems and natural language processing. The Quora dataset comprises over 400,000 question pairs, which are divided into training and test sets. Each pair consists of two questions: an original question and a similar question to the original question. The task is to determine whether the two questions have the same answer based on their similarity in semantics.

MedNLI (Romanov and Shivade, 2018): MedNLI (Medical Natural Language Inference) is a natural language inference dataset developed specifically for the medical field. It was released by researchers from the University of Washington, Stanford University, and MIT. The dataset contains 14,410 sentence pairs, which are divided into training, development, and test sets. Each pair consists of a premise and a hypothesis, both of which are medical texts. The task involves judging whether the hypothesis is entailed by the premise, contradicts the premise, or is neutral, based on the relationship between the premise and the hypothesis.

SST-5 (Socher et al., 2013): The SST-5 dataset is a widely used sentiment classification dataset that consists of 11,855 movie review sentences. It is named after the Stanford Sentiment Treebank, with "5" indicating the 5 sentiment categories: very negative, slightly negative, neutral, slightly positive, and very positive. This dataset was created by researchers at Stanford University and is labeled with one of the 5 sentiment categories, making it a valuable resource for training and evaluating sentiment analysis algorithms.

4.2. Parameter settings

During the model training process, the maximum length of the text was set to 512, and the hidden layer dimension was set to 200. The learning rate, which controls the step size of the model updates during training, was set to 0.00005. The optimizer used to optimize the model parameters during training was Adam. To prevent overfitting, a dropout rate of 0.2 was applied, and an early stopping mechanism was implemented in this study. The model stops training when it fails to improve on the validation set for three consecutive epochs.

Table 1
The accuracy (%) of the model on the Snli, Scitail, Quora, and Sick test sets. Bold black is the best performance of the listed methods.

Model	Snli	Scitail	Quora	Sick
SWEM (Shen et al., 2018)	83.8	–	83.0	–
HBMP (Talman et al., 2018)	86.6	86.0	–	–
DITM (Yu et al., 2021)	–	89.2	86.1	–
DRr-Net (Zhang et al., 2019)	87.7	87.4	89.8	–
BIMPM* (Wang et al., 2017)	87.9	75.3	88.2	76.6
ESIM* (Chen et al., 2017)	88.0	82.4	85.4	76.4
RE2* (Yang et al., 2019)	88.7	86.6	89.4	79.8
DEIM (Jiang et al., 2022b)	88.9	89.5	90.1	80.1
MFAE (Zhang et al., 2020)	90.0	–	90.5	–
BERT_base* (Devlin et al., 2019)	89.6	92.0	89.9	85.4
GPT (Radford et al., 2018)	89.9	88.3	–	–
KIM* (Chen et al., 2018)	88.6	83.1	86.5	78.2
KETM* (Jiang et al., 2023)	89.9	92.3	90.1	86.5
ExBert* (Gajbhiye et al., 2021)	90.5	92.4	90.7	86.8
DIKE-KB*	90.2	92.1	90.7	85.8
DIKE*	**90.6**	**92.6**	**91.0**	**87.1**

4.3. Experimental on natural language inference tasks

4.3.1. Experimental results

We conducted experiments on our model on multiple datasets, and the results are shown in Table 1. DIKE-KB represents our model without using external knowledge, that is, the model only has a text encoding module and an output module. "∗" indicates the results we reproduced.

Our experimental results are summarized in Table 1. We report the accuracy of each model on each dataset and highlight the best-performing models in bold. Our results show that the pre-training-based models generally outperform the other models, this indicates that pre-training techniques are very effective in improving the performance of text-matching tasks. With DIKE achieving the highest accuracy on all four datasets. Specifically, DIKE outperforms the second-best model by 0.4–1.3% on different datasets. In addition, the DIKE model performs better when using external knowledge compared to when not using it. This indicates that external knowledge can significantly improve the performance of text-matching tasks.

Compared with representation-based matching models, our results have shown significant improvement. The accuracy of the best-performing models on three datasets (SNLI, Scitail, and Quora) has increased by 4.0, 6.6, and 8.0 percentage points, respectively. They are slightly inferior to other interaction-based and pre-trained models. This suggests that in text-matching tasks, capturing deeper interactions between texts may not be possible solely through information in the representation layer.

Compared to interaction-based models, our model achieved the highest accuracy among the listed methods, with improvements of 0.7, 3.1, 0.9, and 7.0 percentage points on the four datasets, respectively, compared to the high-accuracy DEIM model. This indicates that our method also outperforms these models in terms of interaction.

Compared with the pre-trained language models BERT_base, GPT, and the MFAE model that uses BERT as the encoder, our model also achieves the best performance on four datasets. This indicates that incorporating external knowledge on top of pre-trained models can also improve the performance of the models.

Compared to other knowledge-enhanced models, our DIKE model demonstrates superior performance across various datasets. Specifically, DIKE outperforms KIM, KETM, and ExBert models on the SNLI dataset, with advantages ranging from 0.1% to 0.7%. On the Scitail and Quora datasets, DIKE also surpasses other knowledge-enhanced models, with accuracy improvements of approximately 0.4% and 0.2%, respectively. Furthermore, DIKE exhibits significant performance enhancements when leveraging external knowledge sources, highlighting its advantage in knowledge enhancement. Compared to representation-based and interaction-based models, DIKE demonstrates notable accuracy improvements, indicating its ability to capture deeper semantic relationships and more refined text interactions.

In summary, our experimental results show that our proposed text-matching model, DIKE, performs well on different datasets and has better generalization ability than other models based on representation, interaction, and pre-training. At the same time, different types of text-matching models perform differently on different datasets, indicating that it is necessary to choose a suitable model based on the specific situation in practical applications.

4.3.2. Experimental results and analysis of different models

To validate the general applicability of our proposed knowledge enhancement method, we conducted experiments using two different models, the ESIM model which does not utilize the Transformer architecture and the BERT model which does. By incorporating external knowledge on top of these models, we achieved improved performance on four different datasets. As shown in Fig. 5, on the ESIM model, we observed accuracy improvements of 0.7%, 3.1%, 2.9%, and 4.4% on the four datasets, respectively. These results demonstrate the effectiveness of our knowledge enhancement framework in improving the performance of text-matching models based on non-Transformer architectures. Similarly, on the BERT model, we observed accuracy improvements

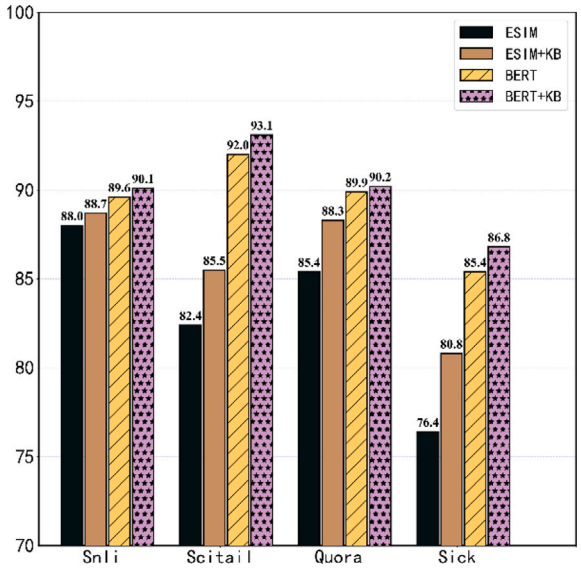


Fig. 5. Experimental results for different training set sizes on the Snli dataset.

of 0.5%, 1.1%, 0.3%, and 1.4% on the four datasets, respectively. These results demonstrate the effectiveness of our knowledge enhancement framework in improving the performance of text-matching models based on Transformer architectures. In summary, our experimental results show that incorporating external knowledge on top of pre-trained models can improve the performance of text-matching models on different datasets, which indicates the potential of our knowledge enhancement framework for practical applications.

4.3.3. Experimental results and analysis of training sets of different sizes

To investigate the role of knowledge in different sizes of training datasets, we trained our model on a portion of the SNLI dataset's training set. The results, as shown in Fig. 6, demonstrate that the improved model achieved higher accuracy than the original model across all training data sizes. This indicates that our improved model is more scalable and flexible, making it capable of adapting to datasets of varying scale and complexity. Furthermore, our results highlight the significant impact of training data on model performance, as increasing the amount of training data led to improvements in accuracy. Additionally, we found that the inclusion of external knowledge had a substantial impact on our model's performance when trained on only 1% of the data. Specifically, external knowledge contributed greatly to the model's accuracy in cases where data was limited. When we trained our model on 10%, 50%, and 100% of the training data, we observed that the incorporation of knowledge led to accuracy improvements of 1.6, 0.9, and 0.6 percentage points, respectively. These experimental results suggest that external knowledge plays a promoting role across datasets of different sizes and that its effect is more pronounced with smaller training datasets.

In conclusion, our comprehensive experimental outcomes validate the effectiveness of the approach we introduce. We determine that the refined model consistently surpasses the baseline across datasets of various scales, further corroborating the effectiveness and superior performance of our model.

4.3.4. Experimental results for the adversarial dataset

To verify the role of knowledge on adversarial datasets, we trained on the SNLI training set and tested on the adversarial BreakNLI dataset. The BreakNLI dataset is primarily designed to test a model's inference knowledge of vocabulary words, with the premise of the test set being constructed from sentences in the SNLI training set. Each sentence is assumed to have one word replaced to obtain a new text pair. The sample table, shown in Table 2, displays entailment examples generated by replacing words with synonyms or hypernyms (i.e., words that are more general than the original word), contradiction examples generated by replacing words with antonyms, and neutral examples generated by replacing words with hyponyms (i.e., words that are more specific than the original word). Our experimental results are presented in Table 3. As shown in Table 3, the ESIM, BIMPM, and RE2 models, which performed well on the SNLI dataset, exhibited a significant decrease in performance on the BreakNLI dataset. Incorporating the external knowledge selected in this paper into the ESIM model increased its accuracy by 13 percentage points. Moreover, adding external knowledge improved the performance by 3.5 percentage points on the adversarial dataset compared to the model proposed in this paper. The primary reason for this improvement may be that our method provides word explanations, which allows the model to infer relationships between words based on their explanations, thereby performing well on the adversarial dataset BreakNLI. The experimental results also indicate that adding external knowledge is effective for our model.

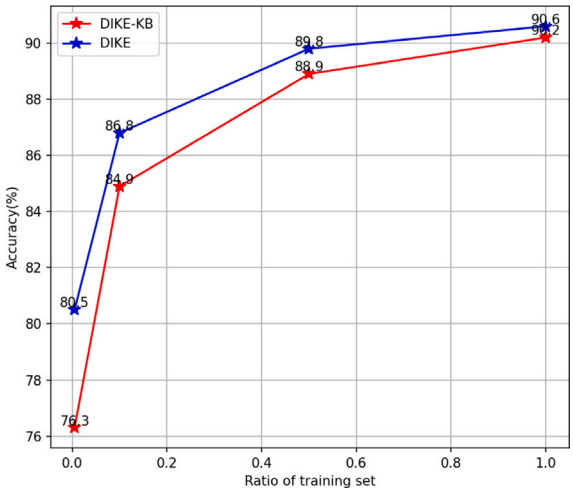


Fig. 6. Experimental results for different sizes of training data.

Table 2
BreakNLI dataset sample table.

Premise	Hyponyms	Label
The cat is sleeping on the couch	The cat is resting on the couch	Entailment
The cat is sleeping on the couch	The pillow is on the couch	Neutral
The cat is sleeping on the couch	The cat is awake on the couch	Contradiction

Table 3
The accuracy (%) of the model on the BreakNLI dataset. Bold black is the best performance of the listed methods.

Model	BreakNLI
BIMPM* (Wang et al., 2017)	68.3
RE2* (Yang et al., 2019)	80.9
KIM (Chen et al., 2018)	83.8
ESIM (Chen et al., 2017)	65.8
ESIM+KB*	78.8
DIKE-KB*	87.7
DIKE*	**91.2**

Table 4
Impact of different knowledge on modeling on SICK, Scitail datasets.

Model	Sick	Scitail
Model(N)	86.0	91.7
Model(C)	86.5	92.1
Model(W)	87.1	92.3
Model(C+W)	**87.8**	**92.6**

4.3.5. Experimental results for different knowledge types

To evaluate the impact of external knowledge, experiments were conducted on the validation sets of SICK and Scitail, with results presented in Table 4. Model(N) refers to the model without any external knowledge, Model(C) incorporates only ConceptNet triplet knowledge, Model(W) incorporates only Wiktionary triplet knowledge, and Model(C+W) incorporates both. The absence of external knowledge led to a decrease in accuracy by 1.8% and 0.9% on the respective datasets, highlighting its significant contribution. Notably, knowledge from Wiktionary was more effective in enhancing model performance than traditional structured data from triplet datasets, likely due to Wiktionary's detailed word explanations improving understanding of complex word relationships. The experiments confirm the efficacy of the proposed model and emphasize the crucial role of external knowledge in enhancing model performance.

4.3.6. Ablation experiments

To validate the effectiveness of the model fusion layer and the text-matching layer modules, we conducted ablation experiments on the validation sets of Sick and Scitail. When the fusion function was not used, which means directly concatenating the text

Table 5
Results of ablation experiments.

	SICK	Scitail
Full model	87.0	92.2
w/o fusion	85.4	91.1
w/o bi-att	86.0	91.6

P: The blonds girl is surfing
H: A blond girl is riding the waves
KB: 'surfing' : The pastime or sport of riding waves on a surfboard
w/o KB:1
w/ KB:0
label:0

Fig. 7. Example from SICK dataset showing the effectiveness of entity descriptions.

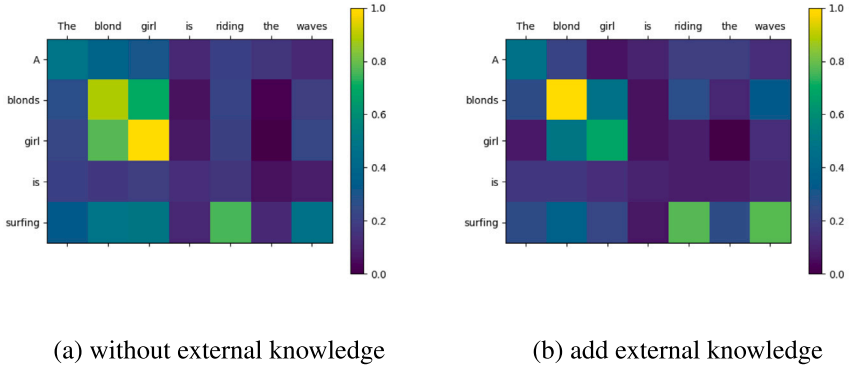


Fig. 8. Visualized attention weight diagram, vertical and horizontal axes represent premises and hypotheses respectively, colors indicate their weights, no external knowledge added to the left, external knowledge added to the right.

information with the knowledge information, the results are shown in Table 5. The table reveals that the accuracy decreased by 1.6 and 1 percentage point, respectively, indicating that the fusion function we proposed can eliminate the noise introduced by knowledge to some extent. Moreover, not using bidirectional attention in the text-matching layer resulted in a decrease of 1 and 0.5 percentage points, respectively. This shows that performing bidirectional attention interaction after the encoding stage can better fuse the information of sentence pairs.

4.3.7. Case study

We randomly selected several examples to be included in the model for prediction. The premise we entered was "The blonde girl is surfing" and the hypothesis was "A blonde girl is riding the waves". The results are shown in Fig. 7. From Fig. 7, we can see that when there is no external knowledge, the model can only infer that the two sentences are not related, resulting in an incorrect judgment. However, after incorporating external knowledge, the model can correctly deduce the entailment relationship between the two sentences, proving the effectiveness of our method.

We visualized the results in Fig. 8, where Fig. 8(a) shows the attention heatmap without external knowledge, and Fig. 8(b) shows the attention heatmap with external knowledge. Without external knowledge, the similarity between "surfing" and "riding waves" is small, so the model judges them to be neutral. However, with external knowledge, the explanation of "surfing" in the external knowledge includes the meaning of surfing on a surfboard, which can be used to link "riding waves", thereby increasing the similarity between the two and allowing the model to correctly infer the entailment relationship. This indicates that the introduction of external knowledge can improve the model's prediction accuracy to some extent.

However, in some cases, incorporating external knowledge may not be effective, as shown in Fig. 9. When the premise is "A girl in white is dancing", and the hypothesis is "A girl is wearing white clothes and is dancing", the definition of "white" is "Bright and colorless; reflecting equal quantities of all frequencies of visible light". Even with external knowledge, we cannot determine that the meaning of "white" in the premise refers to "white clothes", and the attention weight between them will not increase. Our external knowledge only explains "white" and is not related to the context, indicating that the external knowledge we incorporate has certain limitations in some aspects.

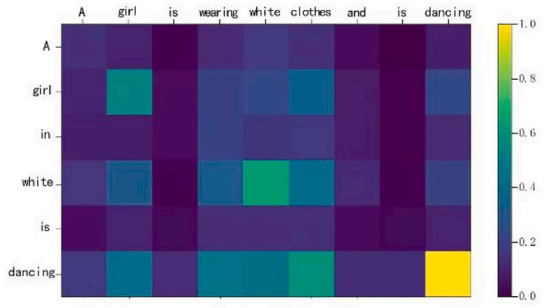


Fig. 9. Error case attention weight.

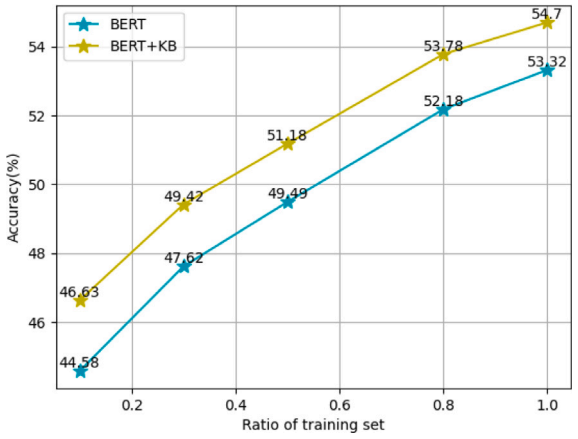


Fig. 10. Experimental results for different training set sizes on the SST-5 dataset.

Table 6
The accuracy (%) of the model on the SST-5 dataset. Bold black is the best performance of the listed methods.

Model	SST-5
Emo2Vec (Xu et al., 2018)	41.6
SWEM-concat (Shen et al., 2018)	46.1
Star-Transformer (Guo et al., 2019)	53.0
BP-Transformer (Ye et al., 2019)	52.7
BERT-PT (Xu et al., 2019)	53.2
BERT (Devlin et al., 2019)	53.3
BERT+KB(ours)*	**54.7**

4.4. Experimental on text classification tasks

4.4.1. Experimental results

The experimental results on the SST-5 text classification dataset in this study are presented in Table 6. As shown in Table 2, our proposed method achieved an accuracy of 54.7% on the SST-5 dataset, outperforming the other listed methods and demonstrating its effectiveness. Moreover, the inclusion of external knowledge improved the model's performance by 1.4 percentage points when compared to the model without external knowledge, indicating its promoting effect. Our model also outperformed the BERT-PT model with a 1.5 percentage point improvement, indicating its superior generalization ability.

4.4.2. Experimental results for different training set sizes on the SST-5 dataset

To examine the impact of knowledge-enhanced frameworks on varying sizes of training data, we trained a model on a subset of the SST-5 dataset. The results, depicted in Fig. 10, show that incorporating external knowledge improved the accuracy by 2.1, 1.8, 1.69, 1.60, and 1.38 percentage points when trained on 10%, 30%, 50%, 80%, and 100% of the training data, respectively. Our experimental findings indicate that the knowledge-enhanced framework facilitated performance across different training data sizes and exhibited superior efficacy when applied to smaller datasets.

Table 7
Experimental results of Mednli dataset.

Model	Acc(%)
Elmo	71.4
Infersent	73.5
ESIM	73.1
BERT	77.5
BioBERT	80.7
SciBERT	80.9
PubmedBERT	84.2
BioELECTRA	86.3
DIKE	78.6
DIKE(BioBERT)	82.3
DIKE(SciBERT)	82.2
DIKE(PubmedBERT)	84.6
DIKE(BioELECTRA)	**86.7**

PAST MEDICAL HISTORY:Type 2 diabetes mellitus (A)

A. The patient has type 2 diabetes (0.9314)

B. The patient has type 1 diabetes (0.0182)

C. The patient is on insulin (0.0504)

Patient also required multiple pressors to maintain her blood pressure (C)

A. the patient is in septic shock (0.0268)

B. the patient is hypertensive (0.0215)

C. the patient is hypotensive (0.9517)

Fig. 11. Sample analysis table.

4.5. Experimental on Mednli tasks

4.5.1. Experimental results

To demonstrate the effectiveness of our proposed method, we validated the proposed model on the MedNLI dataset by employing different pre-trained language models for word embeddings. The resulting outcomes are presented in Table 7. The DIKE(BioBERT) indicates that we replaced BERT with BioBERT in the text encoding module. We augmented the BioELECTRA model with external knowledge and obtained a current best accuracy of 0.867 on the MedNLI test set. The experimental results in Table 7 revealed several observations: (1) Traditional text matching models such as ESIM and Infersent performed less effectively on the MedNLI dataset, which is specific to the medical field, compared to the pre-trained language models such as BERT. This observation highlights the benefits of pre-trained language models in this context. (2) The current mainstream text-matching models did not perform exceptionally well on the MedNLI dataset as they are less adept at learning medical-related knowledge. However, incorporating medical-related knowledge during pre-training improves the model's performance in the medical field, as demonstrated by the improved performance observed on a pre-trained model generated with medical corpus during secondary training. (3) The effectiveness of our proposed method was demonstrated by the performance improvements in different pre-trained language models such as BERT and BioBERT, after incorporating word definitions.

4.5.2. Case study

When we input a patient's medical history and three possible diagnoses into our model, the examples shown in Fig. 11 demonstrate our model's results, with the corresponding probability of each option in parentheses. When we input "the patient needs multiple pressors to maintain blood pressure" into the model, it can correctly select the right answer from the options, indicating that the model can infer that the patient has hypotension and select the correct answer, assisting doctors in making diagnoses. The experimental results show that our proposed method can improve the accuracy of the model and select the correct answer from the given examples, demonstrating the effectiveness and practicality of our method in the medical field.

5. Conclusion

This paper presents a study on text-matching methods, proposing an approach that incorporates external knowledge to enhance the model. The word meanings and knowledge graph triplets are integrated into the model, and a gating mechanism is used to fuse them with the original text to mitigate noise from knowledge incorporation. The proposed architecture is highly versatile, as the

text encoding module can serve as either a text matching or text classification model. Experimental evaluations were conducted on text-matching datasets, including Snli, Quora, Sick, and Scitail, as well as the text classification dataset SST-5, and the medical inference dataset MedNLI. The results of the experiments demonstrate the effectiveness of our proposed approach.

CRediT authorship contribution statement

Kexin Jiang: Methodology, Writing – original draft, Writing – review & editing. **Guozhe Jin:** Writing – original draft. **Zhenguo Zhang:** Funding acquisition, Writing – review & editing. **Rongyi Cui:** Funding acquisition, Writing – review & editing. **Yahui Zhao:** Funding acquisition, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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